

```

=====
QUESTION 2: Git Repository and Version Control
=====git status= FIN
L STATUS ===== --decorateint=====
root/Yashansh-GLA-CourseRegistration
backup course_info.txt docs src
tudent Name: Yashansh Srivastava
oll Number: 2415001905
ubject Name: Application Build and Release Automation Lab
roject Name: Yashansh-GLA-CourseRegistration
emester: 5th

— backup
— course_info.txt
— docs
— src

directories, 1 file
int: Using 'master' as the name for the initial branch. This default branch
ame
int: is subject to change. To configure the initial branch name to use in al
int: of your new repositories, which will suppress this warning, call:
int:
int:   git config --global init.defaultBranch <name>
int:
int: Names commonly chosen instead of 'master' are 'main', 'trunk' and
int: 'development'. The just-created branch can be renamed via this command:
int:
int:   git branch -m <name>
nitialized empty Git repository in /root/Yashansh-GLA-CourseRegistration/.gi
/
n branch master

o commits yet

ntracked files:
(use "git add <file>..." to include in what will be committed)
    course_info.txt

```

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    course_info.txt

nothing added to commit but untracked files present (use "git add" to track)
[master (root-commit) 2ff397d] Initial project setup
 1 file changed, 5 insertions(+)
 create mode 100644 course_info.txt
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
diff --git a/course_info.txt b/course_info.txt
index c9b2789..2bce9de 100644
--- a/course_info.txt
+++ b/course_info.txt
@@ -3,3 +3,5 @@ Roll Number: 2415001905
 Subject Name: Application Build and Release Automation Lab
 Project Name: Yashansh-GLA-CourseRegistration
 Semester: 5th
+Course/Program Name: B.Tech Computer Science and Engineering
+Project Description: Online Course Registration System for managing and searching university courses.
[master a78627f] Update course information
 1 file changed, 2 insertions(+)
a78627f (HEAD -> master) Update course information
2ff397d Initial project setup
Switched to branch 'course-search'
[course-search 703138b] Add course search module
 1 file changed, 12 insertions(+)
 create mode 100644 src/course_search.txt
* course-search
  main
course-search
Switched to branch 'main'
Files on main before feature merge:
ls: cannot access 'src': No such file or directory
```

```
src/course_search.txt | 12 ++++++
1 file changed, 12 insertions(+)
create mode 100644 src/course_search.txt
course_search.txt
Course Search Module

Purpose:
The Course Search Module allows students to search and find available univers
ity courses.

Features:
1. Search courses by course name.
2. Search courses by course code.
3. Search courses by department.
4. Display matching courses.

Module Status: Development
* 703138b (HEAD -> main, course-search) Add course search module
* a78627f Update course information
* 2ff397d Initial project setup
Switched to branch 'course-update'
[course-update 6addeec] Update project description
1 file changed, 1 insertion(+), 1 deletion(-)
Switched to branch 'main'
[main e0b3913] Revise project description
1 file changed, 1 insertion(+), 1 deletion(-)
Auto-merging course_info.txt
CONFLICT (content): Merge conflict in course_info.txt
Automatic merge failed; fix conflicts and then commit the result.
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
Student Name: Yashansh Srivastava
```

```
Roll Number: 2415001905
Subject Name: Application Build and Release Automation Lab
Project Name: Yashansh-GLA-CourseRegistration
Semester: 5th
Course/Program Name: B.Tech Computer Science and Engineering
Project Description: Online Course Registration System developed for GLA University to manage course registration and support student course search.
[main 43d2aa6] Resolve course information merge conflict
===== CURRENT DIRECTORY =====
/root/Yashansh-GLA-CourseRegistration
===== PROJECT STRUCTURE =====
.
├── backup
├── course_info.txt
├── docs
├── src
│   └── course_search.txt
4 directories, 2 files
===== COURSE INFORMATION =====
Student Name: Yashansh Srivastava
Roll Number: 2415001905
Subject Name: Application Build and Release Automation Lab
Project Name: Yashansh-GLA-CourseRegistration
Semester: 5th
Course/Program Name: B.Tech Computer Science and Engineering
Project Description: Online Course Registration System developed for GLA University to manage course registration and support student course search.
===== GIT STATUS =====
On branch main
nothing to commit, working tree clean
===== ALL BRANCHES =====
  course-search
  course-update
* main
===== CURRENT BRANCH =====
main
===== FEATURE FILE =====
Course Search Module

Purpose:
```

Purpose:

The Course Search Module allows students to search and find available university courses.

Features:

1. Search courses by course name.
2. Search courses by course code.
3. Search courses by department.
4. Display matching courses.

Module Status: Development

===== COMPLETE GIT HISTORY =====

```
* 43d2aa6 (HEAD -> main) Resolve course information merge conflict
| \
| * 6addeec (course-update) Update project description
* | e0b3913 Revise project description
| /
* 703138b (course-search) Add course search module
* a78627f Update course information
* 2ff397d Initial project setup
```

===== FINAL STATUS =====

On branch main

nothing to commit, working tree clean

root@LAPTOP-EN23VQC9:~/Yashansh-GLA-CourseRegistration#

root@LAPTOP-EN23VQC9:~/Yashansh-GLA-CourseRegistration# ~

-bash: /root: Is a directory

root@LAPTOP-EN23VQC9:~/Yashansh-GLA-CourseRegistration#

----- QUESTION 1: Linux Environment -----

We are already inside:

/root/Yashansh-GLA-CourseRegistration

Create required directories

mkdir -p src docs backup

Create course_info.txt

cat > course_info.txt <<'EOF'

Student Name: Yashansh Srivastava

Roll Number: 2415001905

Subject Name: Application Build and Release Automation Lab

Project Name: Yashansh-GLA-CourseRegistration

Semester: 5th

EOF

Display current working directory

pwd

Display directory contents

ls

Display course_info.txt

cat course_info.txt

Display project structure

tree 2>/dev/null || find . -maxdepth 2 -print